

AST 6007-01 Spacecraft Dynamics and Control

Final Examination

December 11, 2017 3:00-5:00 PM

Close Books/Close Notes

In the beginning of your answer sheet, sign the honor code: I never received nor gave help regarding this exam.

1. (40) The equations of motion for the Moon-Earth circular restricted three-body problem is given as follows:

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - \omega_0^2 x = -\frac{\mu_E(x-D_E)}{r_E^3} - \frac{\mu_M(x+D_M)}{r_M^3} + \frac{F_x}{m} \\ \ddot{y} + 2\omega_0 \dot{x} - \omega_0^2 y = -\frac{\mu_E y}{r_E^3} - \frac{\mu_M y}{r_M^3} + \frac{F_y}{m} \\ \ddot{z} = -\frac{\mu_E z}{r_E^3} - \frac{\mu_M z}{r_M^3} + \frac{F_z}{m} \end{cases}, \begin{cases} r_E^2 = (x-D_E)^2 + y^2 + z^2 \\ r_M^2 = (x+D_M)^2 + y^2 + z^2 \\ \mu_E = 398,601 \text{ km}^3/\text{sec}^2 \\ \mu_M = 4,887 \text{ km}^3/\text{sec}^2 \\ D_E = 4,674 \text{ km} \\ D_M = 380,073 \text{ km} \\ \omega_0 = 2.66199 \times 10^{-6} \text{ rad/sec} \end{cases}$$

- A. (10) Show that $(x_e = -444,645.6, y_e = 0, z_e = 0)$ is an equilibrium solution with no propulsive forces.
- B. (20) Linearize the equations of motion with respect to the equilibrium solution given in (A).
- C. (10) Discuss the stability of the equilibrium solution given in (A) based on the linearized equation derived in (B).

$$\mathbf{J}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times (\mathbf{J}\boldsymbol{\omega}) = \mathbf{M}$$

$$\dot{\mathbf{R}} = \mathbf{S}(\boldsymbol{\omega})\mathbf{R}$$

2. (30) Consider the following Euler's rotational equations of motion and quaternion kinematic equations:

$$\mathbf{J}\dot{\boldsymbol{\omega}} = -\mathbf{S}(\boldsymbol{\omega})\mathbf{J}\boldsymbol{\omega} + \mathbf{M}$$

$$\dot{\hat{\mathbf{q}}} = \frac{1}{2}\mathbf{S}(\boldsymbol{\omega})\hat{\mathbf{q}} + \frac{1}{2}\mathbf{q}_4\boldsymbol{\omega}, \quad \dot{\mathbf{q}}_4 = -\frac{1}{2}\boldsymbol{\omega}^T \hat{\mathbf{q}}$$

$$\mathbf{H} = \mathbf{J}\boldsymbol{\omega}$$

$$\boldsymbol{\omega} = [\omega_x \ \omega_y \ \omega_z]^T, \quad \mathbf{M} = [M_x \ M_y \ M_z]^T, \quad \mathbf{q} = [q_1 \ q_2 \ q_3 \ q_4]^T = [\hat{\mathbf{q}} \ q_4]^T, \quad \mathbf{S}(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

- A. (10) Determine the nonlinear control law to stabilize the equilibrium $(\boldsymbol{\omega} = [0 \ 0 \ 0]^T, \mathbf{q} = [0 \ 0 \ 0 \ 1]^T)$ based on Lyapunov approach.

- B. (10) Show that if the initial values of the quaternion satisfy $q_1^2(0) + q_2^2(0) + q_3^2(0) + q_4^2(0) = 1$ then $q_1^2(t) + q_2^2(t) + q_3^2(t) + q_4^2(t) = 1$ holds for all $t \geq 0$.

- C. (10) In case that the rotational motion is torque-free and $\mathbf{J} = \text{diag}(A, B, C)$, show that the kinetic energy and the magnitude of angular momentum are conserved.

$$T = \frac{1}{2}\boldsymbol{\omega}^T \mathbf{J} \boldsymbol{\omega} = \frac{1}{2}(J_x \omega_x^2 + \dots)$$

3. (15) The rotation matrix R can be expressed by the quaternion:

$$R(q) = (q_4^2 - \hat{q}^T \hat{q}) I_{3 \times 3} + 2\hat{q}\hat{q}^T - 2q_4 S(\hat{q})$$

When the orientation is slightly changed as the small angle $\Delta\phi$ about the axis $\mathbf{a} = [a_1 \ a_2 \ a_3]^T$ from the reference frame, derive the rotation matrix R of the infinitesimal rotation as a function of $\Delta\phi$ and $\mathbf{a}$.

Hint: $\sin \phi = 2 \sin \frac{\phi}{2} \cos \frac{\phi}{2}$, $\cos \phi = \cos^2 \frac{\phi}{2} - \sin^2 \frac{\phi}{2}$

4. (15) Does only one quaternion exist for representing one orientation? Justify your answer.

$$\begin{pmatrix} q_4 \\ \mathbf{q} \end{pmatrix} = \begin{pmatrix} \cos \frac{\phi}{2} \\ \mathbf{a} \sin \frac{\phi}{2} \end{pmatrix}$$

$$q_4 = \cos \frac{\phi}{2} ?$$

$$R = e^{S(\hat{\mathbf{a}}) \phi}$$

$$\dot{R} = S(\hat{\boldsymbol{\omega}}) R$$

Ans?

$$\hat{\mathbf{q}} = [q_4 \ \mathbf{q}]$$

$$= \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$